

Project Phase-I Guidelines - BCSE355L - Cloud Architecture Design

Course Instructor: Dr. Priya V

Submission Deadline: 30 July 2026

Abstract

Write a concise abstract (200–300 words) explaining:

Problem statement

Existing challenges

Proposed solution

AWS services used

Literature Survey

Each project must include 15 research papers.

Use papers published in:

- IEEE
- Springer
- Elsevier
- ACM
- Wiley
- MDPI
- Nature
- Scopus Indexed Journals

Prefer papers published between 2023–2026.

Distribution of Papers

Team of 3

Each student studies 5 papers

Student A → Papers 1–5

Student B → Papers 6–10

Student C → Papers 11–15

Team of 2

Student A → Papers 1–8

Student B → Papers 9–15

Individual Project

Single student studies all 15 papers.

Individual Research Gap Analysis

Every student must independently identify:

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- Existing methods
 - Advantages
 - Limitations
 - Research Gap
 - Possible improvement
- Each student should prepare analysis for their assigned five papers.
 - Do NOT copy research gaps directly from the paper.
 - Your analysis should reflect your own understanding.

Suggested Literature Survey Table

Paper	Method	Dataset	Advantages	Limitations	Research Gap
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Project Objectives

Write 4–6 clear objectives.

Example:

Develop an intelligent cloud-based monitoring system.

Reduce response time using AWS services.

Improve prediction accuracy using AI.

Store data securely in the cloud.

Visualize results using dashboards.

Objectives should be measurable and achievable.

Novelty Summary

Students must clearly explain:

What makes your project different from existing work?

Include:

- New feature
- Better algorithm
- Better architecture
- Better AWS integration
- Better security
- Better automation
- Better accuracy

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- Better scalability

Limit to one page.

Proposed Architecture

Two architecture diagrams are mandatory.

Diagram 1- AWS Cloud Architecture

This diagram should explain how AWS services interact.

The architecture should clearly show:

Data flow

Storage

Processing

Authentication

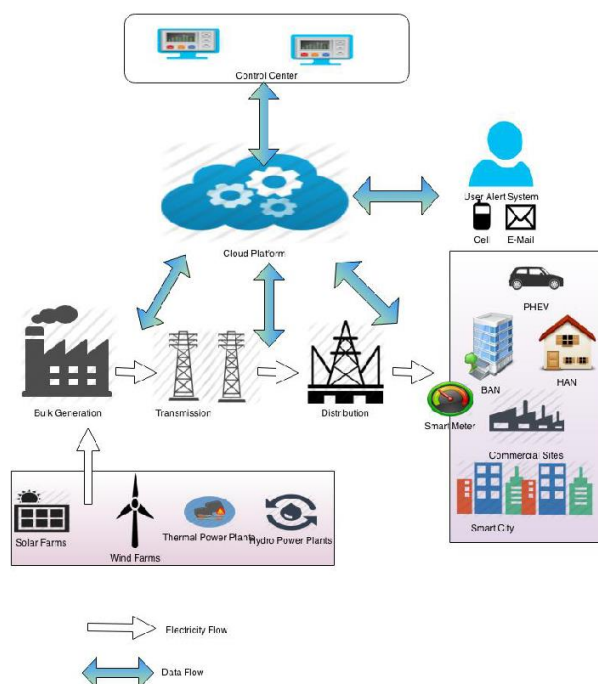
Notifications

Monitoring

Reference - <https://docs.aws.amazon.com/solutions/ev-digital-twin-ai-powered-operational-monitoring>

Diagram 2 - Complete System Architecture

This diagram should explain the overall project workflow.



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Dataset Details

Clearly mention:

- Dataset Name
- Source
- URL
- Size
- Number of Records
- Number of Features
- Data Type
- License
- Purpose of using the dataset
- Preprocessing required

AWS Services Planning

Mention all AWS services planned for implementation.

Example for a Healthcare AI Project

AWS Service	Purpose
Amazon EC2	Host the web application
Amazon S3	Store patient images and datasets
Amazon RDS	Store patient records
AWS Lambda	Process uploaded medical images
Amazon SageMaker	Train and deploy disease prediction model
Amazon API Gateway	Provide REST APIs
AWS IAM	Secure user access
Amazon Cognito	Authenticate doctors and patients
Amazon CloudWatch	Monitor application health
Amazon SNS	Send diagnosis notifications
Amazon QuickSight	Visualize patient statistics

GitHub Guidelines

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Each team must create one GitHub repository.

Repository naming format:

ProjectName_Cloud_Project_2026

Example

SmartFloodMonitoring_Cloud_Project_2026

Sample GitHub repository structure with task allocation for a team of three students.

Smart_Agriculture_Cloud_Project_2026/

```
|
├── README.md
├── LICENSE
├── .gitignore
├── docs/
│   ├── Project_Report.docx
│   ├── Literature_Survey.docx
│   ├── Research_Gap.docx
│   ├── Objectives.docx
│   └── Novelty.docx
├── architecture/
│   ├── AWS_Architecture.png
│   ├── System_Architecture.png
│   └── Workflow.png
├── dataset/
│   ├── raw/
│   ├── processed/
│   └── dataset_description.pdf
├── src/
│   ├── frontend/
│   ├── backend/
│   ├── ml_model/
│   └── aws/
├── results/
└── presentation/
```

GitHub Branch Allocation

```
main
├── develop
```

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```
|
├── feature/student1
├── feature/student2
└── feature/student3
```

Each student works only in their own feature branch.

feature/student2

```
|
├── src/
│   ├── backend/
│   │   ├── APIs
│   │   ├── Controllers
│   │   ├── Database
│   │   └── Authentication
│   └── aws/
│       ├── Lambda
│       ├── S3
│       ├── EC2
│       ├── IAM
│       └── CloudWatch
├── docs/
│   └── Research_Gap_Student2.docx
└── architecture/
    └── AWS_Architecture.png
```

feature/student3

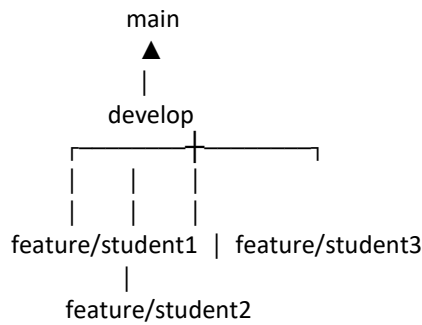
```
|
├── dataset/
│   ├── raw/
│   ├── processed/
│   └── Dataset_Details.docx
├── src/
│   ├── ml_model/
│   │   ├── preprocessing.py
│   │   ├── train.py
│   │   ├── predict.py
│   │   └── model.pkl
├── results/
│   ├── graphs/
│   ├── screenshots/
│   └── accuracy.xlsx
└── docs/
    └── Research_Gap_Student3.docx
```

Final Merge Process

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Workflow

- Create the repository.
- Create the develop branch.
- Each student creates their own feature/studentX branch.
- Students commit their work regularly to their branch.
- Raise a Pull Request (PR) from the feature branch to develop.
- Team members review the changes.
- Resolve any merge conflicts.
- Merge approved changes into develop.
- Once the project is stable, merge develop into main.
- Tag the release (e.g., v1.0-Phase1).

Expected Contribution Matrix - Sample

Activity	Student 1	Student 2	Student 3
Literature Survey (5 Papers)	✓	✓	✓
Research Gap (5 Papers)	✓	✓	✓
Frontend Development	✓		
Backend Development		✓	
Database Integration		✓	
Azure Cloud Services	✓	✓	✓
AI / Machine Learning			✓
Testing	✓	✓	✓
Documentation	✓	✓	✓
Presentation	✓	✓	✓
GitHub Commits	✓	✓	✓

Expected GitHub Activity

Each student should have approximately:

- **20–30 meaningful commits**

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- **At least 2 Pull Requests**
- **Code review participation**
- **Regular weekly commits**
- **Continuous documentation updates**

Important Instruction

- **Hard copy submission is not required.**
- Prepare the complete project report in **soft copy (PDF or Microsoft Word format)**.
- Ensure that all required sections are completed before the review.
- During the project review, each team must **present the soft copy of the document** along with the architecture diagrams, literature survey, research gap analysis, dataset details, GitHub repository, and implementation progress.
- Students should be ready to explain their individual contributions, GitHub commit history, and Azure/AWS service integration during the review.